**32.** Suppose that $f$ is a differentiable function such that $f(\sin x) = x$ for all $x$. Show that, for $-1 < x < 1$, $f'(x) = \frac{1}{\sqrt{1-x^2}}$.

Let $y = \sin x$, then $\cos x = \sqrt{1-y^2}$

$\Rightarrow f'(\sin x) \cdot \cos x = 1$

$\Rightarrow f'(\sin x) = \frac{1}{\cos x}$

$\Rightarrow f'(y) = \frac{1}{\sqrt{1-y^2}}$

$\Rightarrow f'(x) = \frac{1}{\sqrt{1-x^2}}$ #

